UNIT III

3 Big Data Analytics Life Cycle

Syllabus

*Introduction to Big Data, sources of Big Data, **Data Analytic Lifecycle :** Introduction, Phase 1 : Discovery, Phase 2 : Data Preparation, Phase 3 : Model Planning, Phase 4 : Model Building, Phase 5 : Communication results, Phase 6 : Operation alize.*

Contents

3.1 Introduction to Big Data

SPPU : Aug.-18, May-19, Oct.-19, Dec.-19

- Big data can be defined as very volumes of data available at various sources, in varying degrees of complexity, generated at different speed i.e. velocities and varying degrees of ambiguity, which cannot be processed using traditional technologies, processing methods, algorithms or any commerical off-the-shelf solutions.
- 'Big data' is a term used to describe collection of data that is huge in size and yet growing exponentially with time. In short, such a data is so large and complex that none of the traditional data management tools are able to store it or process it efficiently.
- The processing of big data begins with the raw data that isn't aggregated or organized and is most often impossible to store in the memory of a single computer.
- Big data processing is a set of techniques or programming models to access large-scale data to extract useful information for supporting and providing decisions. Hadoop is the open-source implementation of MapReduce and is widely used for big data processing.

3.1.1 Big Data Requirement

- We can classify **Big Data requirements** based on its five main characteristics :

1. Volume :
   - Size of data to be processed is large-it needs to be broken into manageable chunks.
   - Data needs to be processed in parallel across multiple systems.
   - Data needs to be processed across several program modules simultaneously.
   - Data needs to be processed once and processed to completion due to volumes.
   - Data needs to be processed from any point of failure, since it is extremely large to restart the process from the beginning.
2. Velocity :
   - Data needs to be processed at streaming speeds during data collection.
   - Data needs to be processed for multiple acquisition points.
3. Variety :
   - Data of different formats needs to be processed.
   - Data of different types needs to be processed.
   - Data of different structures needs to be processed.
   - Data from different regions needs to be processed.

4. Ambiguity :
   - Big data is ambiguous by nature due to the lack of relevant metadata and context in many cases. An example is the use of M and F in a sentence, it can mean, respectively, monday and friday, male and female or mother and father.
   - Big data that is within the corporation also exhibits this ambiguity to a lesser degree. For example, employment agreements have standard and custom sections and the latter is ambiguous without the right context.
5. Complexity :
   - Big data complexity needs to use many algorithms to process data quickly and efficiently.
   - Several types of data need multi-pass processing and scalability is extremely important.

3.1.2 Benefits of Big Data Processing

Benefits of big data processing

1. Improved customer service.
2. Business can utilize outside intelligence while taking decision.
3. Reducing maintenance costs.
4. Re-develop your products : Big data can also help you understand how others perceive your products so that you can adapt them or your marketing, if need be.
5. Early identification of risk to the product/services, if any.
6. Better operational efficiency.

3.1.3 Big Data Challenges

- Collecting, storing and processing big data comes with its own set of challenges :
  1. Big data is growing exponentially and existing data management solutions have to be constantly updated to cope with the three Vs.
  2. Organizations do not have enough skilled data professionals who can understand and work with big data and big data tools.

3.1.4 Data Analytical Architecture

- Analytics architecture refers to the systems, protocols and technology used to collect, store and analyze data. Analytics architecture also focuses on multiple layers, starting with data warehouse architecture, which defines how users in an organization can access and interact with data.
- Fig. 3.1.1 shows data analytical architecture.

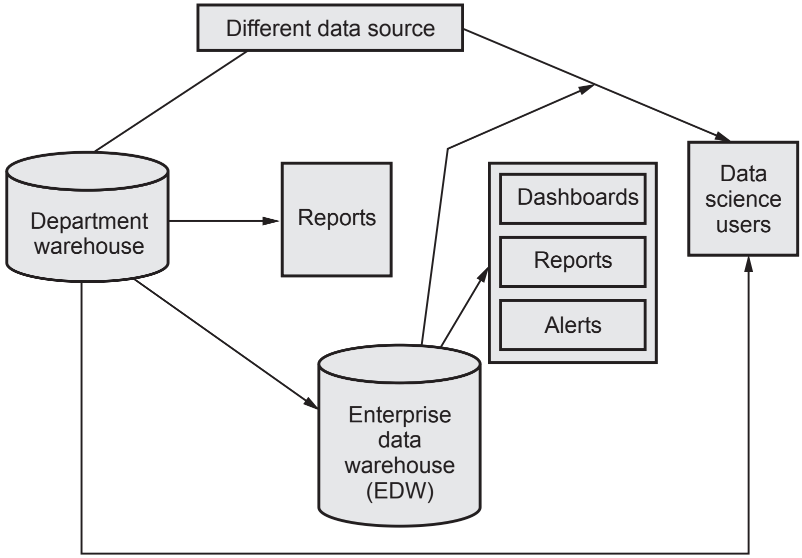

Fig. 3.1.1 Data analytical architecture

- Data to be loaded into the data warehouse. It must be well understood structured and normalized with the appropriate data type. Centralization provides security, backup facility. Also provides significant pre-processing and checkpoints facility before storing data.
- Required level of control on the EDM with additional local systems may emerge in the form of departmental warehouses and local data marts that business users create to accommodate their need for flexible analysis. Sometimes local data marts allow users to do some level of more in-depth analysis.
- Once in the data warehouse, data is read by additional applications across the enterprise for Bl and reporting purposes. These are high-priority operational processes getting critical data feeds from the data warehouses and repositories.
- At last, analysts get data provisioned for their downstream analytics. Many times, these tools are limited to in-memory analytics on desktops analyzing samples of data, rather than the entire population of a dataset.

3.1.5 Big Data Ecosystem

- Big data ecosystem is the comprehension of massive functional components with various enabling tools. Capabilities of the big data ecosystem are not only about computing and storing big data, but also the advantages of its systematic platform and potentials of big data analytics.
- Organizations and data collectors that can gather data from individuals start realizing that a new economy is emerging as a data business. Depending on the evolution of this economy, a new ecosystem is rising.

- Many organizations and data collectors are depend on the data they can gather from individuals which contains great value and, as a result, a new economy is emerging.
- This new digital economy continues to evolve; the market sees the introduction of data vendors and data cleaners that use crowdsourcing to test the outcomes of machine learning techniques.
- As the ecosystem has been growing, groups of interest have been formed. Currently there are four main groups which are associated with each other : Data devices, data collectors, data aggregators and data users and buyers.
- Fig. 3.1.2 shows emerging big data ecosystem.

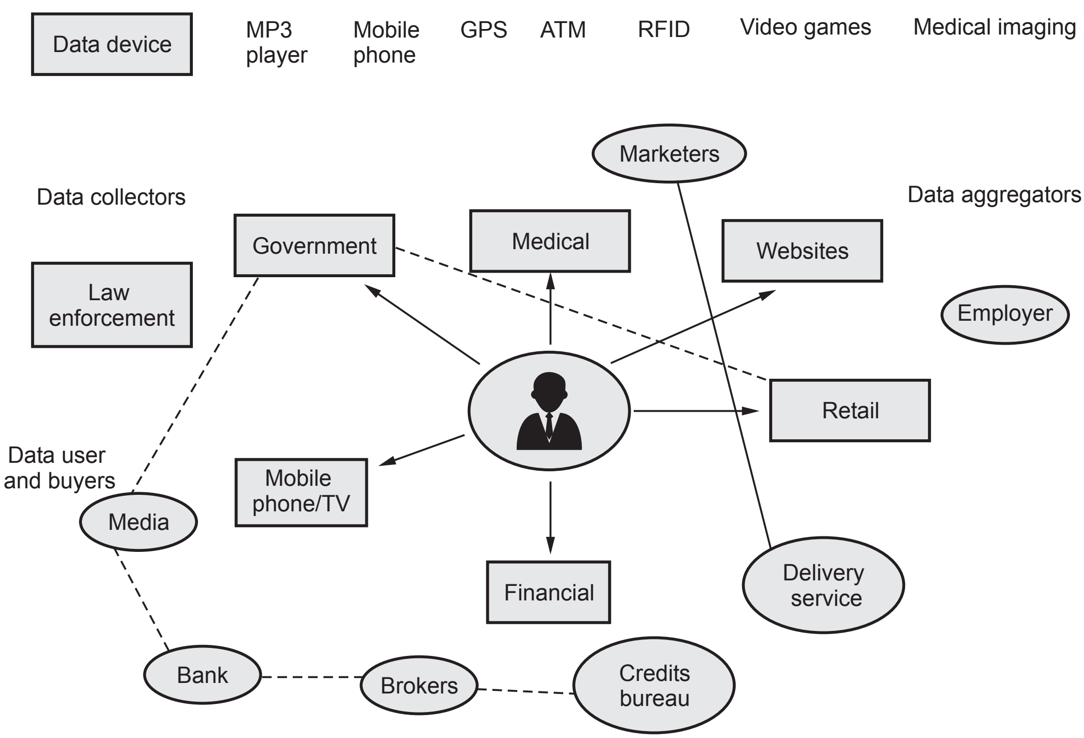

Fig. 3.1.2 Emerging big data ecosystem

1. Data device :

- Data device and sensor network gather data from various locations and continuously generate new data.
- **Example of data devices :** Playing games, smart phone and retail **shopping**.
- **Sensor data :** Growing network of sensor devices generate data based on montoring environmental coditions, such as temperature, sound, pressure, power water levels etc.

- This data can have a wide range of practical application if collected, aggregated, analyzed and acted upon. Examples include, water lavel monitoring and smart home monitoring.
- Mobile networks : Mobile network generates large number of data to share picture, video, audio file and text. These data is process at every mobile tower with associated demographics, location latencies etc. Sometime mobile network may crash for large number of data movement takes place.
- Retail shopping loyalty cards records not just the amount an individual spends, but the location of stores that person visits, the kinds of product purchased, the store where goods are purchased most often and combinations of product purchased together.

2. Data collectors :

- Data collectors : It includes samples entities that collect data from the device and user.
- Retail stores tracking the path a customer takes through their store while pushing a shopping cart with an RFID chip so they can gauge which products get the most foot traffic using geospatial data collected from the RFID chips.
- Data collectors example : Government, retail stores, cable TV provider.
- Cable TV provider which tracks the shows that a person watches.

3. Data aggregators :

- The entities which process collected data from the first layer make them understandable. They give them additional value to prepare them for the handing over process. Now the data is ready to be offered on the market.
- Typically, one of these data aggregators can transform and package the data as products to sell to list brokers that might want to generate marketing lists of people who may be good targets for specific ad compaigns.

4. Data users and buyers :

- The enetities represent a group of the final layer from the big data ecosystem. This group has the final benefits from the collected and aggregated data offered by the data aggregators.
- Data users may wanto to track or prepare for natural disaster by identifying which areas a hurricane will affect first hand. It can be observed by tracking tweets it or discussing it in social media.

Review Questions

1. *What is Big data ? Explain characteristics of Big Data.* **SPPU : Aug.-18 (In Sem), Marks 4**
2. *Explain current analytics architecture with suitable diagram.* **SPPU : Aug.-18 (In Sem), Marks 6**
3. *Explain bigdata ecosystem.* **SPPU : Aug.-18 (In Sem), Marks 4**
4. *What is big data ? Explain 3V's of big data.* **SPPU : May-19 (End Sem), Marks 5**
5. *Draw and explain current analytical architecture.* **SPPU : Oct.-19 (In Sem), Marks 5**
6. *Enlist and explain various users involved to make successful analytical project.* **SPPU Oct.-19 (In Sem), Marks 5**
7. *Discuss with example data devices and data collectors of emerging big data ecosystem.* **SPPU : Oct.-19 (In Sem), Marks 5**
8. *Draw and Explain big data ecosystem.* **SPPU : Dec.-19 (End Sem), Marks 5**
9. *Explain current analytical architecture with diagram.* **SPPU : Dec.-19 (End Sem), Marks 5**

3.2 Sources of Big Data

- Machine data consists of information generated from industrial equipment, real-time data from sensors that track parts and monitor machinery and even web logs that track user behavior online.
- At Arcplan client CERN, the largest particle physics research center in the world, the Large Hadron Collider (LHC) generates 40 terabytes of data every second during experiments.
- Regarding transactional data, large retailers and even B2B companies can generate multitudes of data on a regular basis considering that their transactions consist of one or many items, product IDs, prices, payment information, manufacturer and distributor data and much more.
- Some of the examples of big data are :
  1. **Social media :** Social media is one of the biggest contributors to the flood of data we have today. Facebook generates around 500+ terabytes of data everyday in the form of content generated by the users like status messages, photos and video uploads, messages, comments etc.
  2. **Stock exchange :** Data generated by stock exchanges is also in terabytes per day. Most of this data is the trade data of users and companies.
  3. **Aviation industry :** A single jet engine can generate around 10 terabytes of data during a 30 minute flight.
  4. **Survey data :** Online or offline surveys conducted on various topics which typically have hundreds and thousands of responses and need to be processed

for analysis and visualization by creating a cluster of population and their associated responses.

5. **Compliance data :** Many organizations like healthcare, hospitals, life sciences, finance etc, has to file compliance reports.

3.2.1 Data Repository

- Data repository is also known as a data library or data archive. This is a general term to refer to a data set isolated to be mined for data reporting and analysis. The data repository is a large database infrastructure, several databases that collect, manage and store data sets for data analysis, sharing and reporting.

  1. Spreadsheets :
     - Spreadsheets enabled business user to crate simple logic on data structured in rows and columns and create their own analyses of business problems.
     - Database administrator training is not required to create spreadsheets : They can be set up to do many things quickly and independently of Information Technology (IT) groups.

  - Spreadsheets are easy to share and en users have control over the logic involved.

  2. Enterprise Data Warehouse (EDWs) are critical for reporting and 81 tasks and solve many of the problems that profilerating spreadsheets introduce, such as which of the multiple versions of a spreadsheet is correct.
     - From an analyst perpective, EDW and 81 solve problems related to data accuracy and availability.
     - But EDW and BI introduce new problems related to flexibility and agility, which were less pronounced when dealing with spreadsheets.
  3. Analytic sandbox : Analytic sandbox is solution to this problem. It attempts to resolve the conflict for analysts and data scientists with EDW and more formally managed corporate data.
  4. In this model, the IT group may strill manage the analytic sandboxes, but they will be purposefully designed to enable robust analytics, while centrally managed and secured.

3.2.2 Example of Data Repository

1. Data warehouse is a large repository that aggregates data usually from multiple sources or segments of a business, without the data being necessarily related.
2. Data lake is a large repository that stores unstructed data that is classified and tagged with metadata.

3. Data marts are subsets of the data repository. These data marts are more targeted to what the data user needs and easier to use.
4. Metadata repositories store data about data and databases. The metadata explains where the data source, how it was captured and what it represents.
5. Data cubes are lists of data with three or more dimensions stored as a table.

3.2.3 Advantages and Disadvantages of Data Repository

Advantages :

1. Data is preserved and archived.
2. Data isolation allows for easier and faster data reporting.
3. Data adminitratrors have easier time tracking problems.
4. There is value to storing and analyzing data.

Disadvantages :

1. Growing data sets could slow down systems.
2. A system crash could affect all the data.
3. Unauthorized users can access all sensitive data more easily than if it was distributed across several locations.

3.2.4 Analytic Sandbox

- An analytical sandbox is used as a tool for data storage manipulation and exploration separated from a production environment due to getting flexibility and freelance that is caused by this separation.
- The policy of accessing a production environment is tightly controlled due to its importance and need for financial reporting.
- An analytical sandbox sometimes mentioned as a workspace.
- An analytcial sandbox is primarily assigned for data analysts who require access to al data throughout the organization.
- Analytic sandbox are designed to enable teams to explore many datasets in a controlled fashion and are not typically used for enterprise level finanical reporting ans sales dashboards.
- Many times, analytics sandboxes enable high-performance computing using in-database processing, the analytics occur within the database itself.

- Analytical sandboxes attempt to provide a less restricted environment to enable robust analytics being centrally managed and secured.
- Rather than the typical structured data in the EDW, analystic sandboxes can house a greater variety of data, such as raw data, textual data and other kinds of unstructured data, without interfering with critical production databases.

3.2.5 Factor Responsible for Data Volume in Big Data

1. **Machine data :** Machine data contains a definitive record of all activity and behavior of your customers, users, transitions, applications, servers, networks, factory machinery and so on. It's configuration data, data from APIs and message queues, change events, the output of diagnostic commands and call detail records, sensor data from remote equipment and more.
2. **Application log :** Most homegrown and packaged applications write local logfiles, logging services built into application severs like Web Logic, Web Sphere and JBoss. These files are critical for day-to-day debugging of production applications by developers and application support. When developers put timing information into their log events, they can also be used to monitor and report on application performance.
3. **Business process logs :** Complex events processing and business process management system logs are treasure troves of business and IT relevant data. These logs are will generally include definitive records of customer activity across multiple channels such as the web, IVR/contact center or retail.
4. **Clickstream data :** User activity on the Internet is captured in clickstrem data. This provides insight into a user's website and web page activity. The informations is valueble for usability analysis, marketing and general research.
5. **Thrid party data :** The sensititve data that's not in databases is on file systems. In some industries such as healthcare, the biggest data leakage risk is consumer records on shared file systems. Different OS, third-party tools and storage technologies provide different options for auditing read access to sensitive data at the file systems. Different options for auditing read access to sensitive data at the file system level. This audit data is a vital data source for monitoring and investigating access to sensitive data.
6. **Electronic mails :** Every company have large collection of emails generated by customers, employees and executives on daily basis. These email communications are an important asset to an organization, which are audited case-by case basis and entire life cycle management of emails is done.

3.3 Data Analytic Lifecycle

SPPU : Aug.-18, Dec.-18, May-19, Oct.-19

- The data analytic lifecycle is designed for big data problems and data science projects. With six phases the project work can occur in several phases simultaneously. The cycle is iterative to portray a real project. Work can return to earlier phases as new information is uncovered.
- According to Dietrich (2013), it is a cyclical life cycle that has iterative parts in each of its six steps :

1) Discovery	2) Pre-processing data
3) Model planning	4) Model building
5) Communicate results	6) Operationalize

- Fig. 3.3.1 shows data analytic lifecycle.

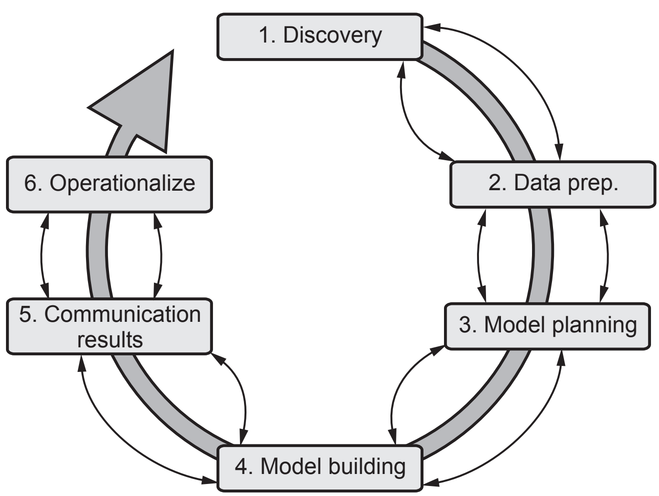

Fig. 3.3.1 Data analytic life cycle

3.3.1 Phase 1 : Discovery

- This phase is all about defining the data's purpose and how to achieve it by the end of the data analytics lifecycle. The stage consists of identifying critical objectives a business is trying to discover by mapping out the data.
- During this process, the team learns about the business domain and checks whether the business unit or organization has worked on similar projects to refer to any learnings.
- In this phase, the team also evaluates technology, people, data and time. For example, while dealing with a small dataset, the team can use excel.

- Phase 1 contains following process :
  a) Learning the business domain
  b) Resources
  c) Framing the problem
  d) Identifying key stakeholders
  e) Interviewing the analytics sponsor
  f) Developing initial hypotheses
  g) Identifying potential data sources

3.3.2 Phase 2 : Data Preparation

- This stage involves collecting, processing and cleaning data. Here the focus shifts from business requirements to data requirements. In this early phase, data is collected but not analyzed.
- The data preparation phase is generally the most iterative and the one that teams tend to underestimate most often.

a) Preparing the analytic sandbox :

- Create the analytic sandbox. It also called a workspace. It allows the team to explore data without interfering with live production data.
- Sandbox collects all kinds of data. The sandbox allows organizations to undertake ambitious projects beyond traditional data analysis and BI to perform advanced predictive analytics.

b) Performing ETLT (Extract, Transform, Load, Transform) :

- The team needs to execute Extract, Load and Transform (ELT) to get data into the sandbox.
- Extract, Transform, Load (ETL) : It transforms the data based on a set of business rules before loading it into the sandbox.
- Extract, Load, Transform (ELT) : It loads the data into the sandbox and then transforms it based on a set of business rules.
- Extract, Transform, Load, Transform (ETLT) : It's the combination of ETL and ELT and has two transformation levels.

c) Learning about the data :

- Data is captured through three main ways :
  i. Data acquisition : Obtaining existing data from outside sources.
  ii. Data entry : Creating new data values from data inputted within the organization.

iii. Signal reception : Capturing data created by devices.

d) Data Conditioning :

- Data conditioning includes cleaning data, normalizing datasets and performing transformations. It is often viewed as a preprocessing step prior to data analysis; it might be performed by data owner, IT department, DBA, etc.
- Best to have data scientists involved and data science teams prefer more data than too little.

e) Common tools for data preparation :

- Hadoop can perform parallel ingest and analysis.
- Alpine miner provides a graphical user interface for creating analytic workflows.
- OpenRefine is a free, open source tool for working with messy data.
- Similar to OpenRefine, data wrangler is an interactive tool for data cleansing and transformation.

3.3.3 Phase 3 : Model Planning

- The team determines the methods, techniques and workflow it intends to follow for the subsequent model building phase.
- The team explores the data to learn about the relationships between variables and subsequently selects key variables and the most suitable models.
- Activities to consider :
  a) Assess the structure of the data - this dictates the tools and analytic techniques for the next phase.
  b) Ensure the analytic techniques enable the team to meet the business objectives and accept or reject the working hypotheses.
  c) Determine if the situation warrants a single model or a series of techniques as part of a larger analytic workflow.
  d) Research and understand how other analysts have approached this kind or similar kind of problem.

a) Data exploration and variable selection

- Explore the data to understand the relationships among the variables to inform selection of the variables and methods. A common way to do this is to use data visualization tools.

- Often, stakeholders and subject matter experts may have ideas. For example, some hypothesis that led to the project.
- Aim for capturing the most essential predictors and variables. This often requires iterations and testing to identify key variables.
- If the team plans to run regression analysis, identify the candidate predictors and outcome variables of the model.

b) Model selection

- The main goal is to choose an analytical technique or several candidates, based on the end goal of the project. We observe events in the real world and attempt to construct models that emulate this behavior with a set of rules and conditions.
- A model is simply an abstraction from reality. Determine whether to use techniques best suited for structured data, unstructured data or a hybrid approach.
- Teams often create initial models using statistical software packages such as R, SAS or Matlab. Which may have limitations when applied to very large datasets.
- The team moves to the model building phase once it has a good idea about the type of model to try.

c) Common tools for the model planning phase

- R programming language has a complete set of modeling capabilities. It contains about 5000 packages for data analysis and graphical presentation.
- SQL analysis services can perform in-database analytics of common data mining functions, involved aggregations and basic predictive models.
- SAS/ACCESS provides integration between SAS and the analytics sandbox via multiple data connections.

3.3.4 Phase 4 : Model Building

- The team develops datasets for testing, training and production purposes. In addition, in this phase the team builds and executes models based on the work done in the model planning phase.
- Building a model involves two phases :
  - **a) Design the model :** Identify a suitable model. This step can involve a number of different modeling techniques to identify a suitable model. These may include decision trees, regression techniques and neural networks.
  - **b) Execute the model :** The model is run against the data to ensure that the model fits the data.
- Common commercial tools for the model building phase :
  - a. SAS enterprise miner used for building enterprise-level computing and analytics.

b. SPSS modeler (IBM) provides enterprise-level computing and analytics.

c. Matlab is a high-level language for data analytics, algorithms, data exploration.

d. Alpine miner provides GUI frontend for backend analytics tools.

e. STATISTICA and MATHEMATICA is popular data mining and analytics tools.

3.3.5 Phase 5 : Communicate Results

- This phase aims to determine whether the project results are a success or failure and start collaborating with significant stakeholders.
- The team identifies the vital findings of their analysis, measures the associated business value and creates a summarized narrative to convey the stakeholders' results.
- Communicate and document the key findings and major insights derived from the analysis. This is the most visible portion of the process to the outside stakeholders and sponsors.

3.3.6 Phase 6 : Operationalize

- This final phase moves data from the sandbox into a live environment. Data is monitored and analyzed to see if the generated model is creating the expected results. If the results aren't as expected, you can return to any of the preceding phases to tweak the data.
- The team communicates the benefits of the project more broadly and sets up a pilot project to deploy the work in a controlled way. Risk is managed effectively by undertaking small scope, pilot deployment before a wide-scale rollout.
- During the pilot project, the team may need to execute the algorithm more efficiently in the database rather than with in-memory tools like R, especially with larger datasets.
- To test the model in a live setting, consider running the model in a production environment for a discrete set of products or a single line of business. Monitor model accuracy and retrain the model if necessary.
- Key outputs from successful analytics project

  a) Business user tries to determine business benefits and implications.

  b) Project sponsor wants business impact, risks, ROI.

  c) Project manager needs to determine if project completed on time, within budget, goals met.

  d) Business intelligence analyst needs to know if reports and dashboards will be impacted and need to change.

e) Data engineer and DBA must share code and document.

f) Data scientist must share code and explain model to peers, managers, stakeholders.

Review Questions

1. *Explain different phases of data analytics life cycle.* **SPPU : Aug.-18 (In Sem), Marks 6**
2. *Explain data analytic life cycle.* **SPPU : Dec.-18 (End Sem), Marks 8**
3. *Draw data analytics lifecycle and give brief description about all phases.* **SPPU : May-19 (End Sem), Marks 5**
4. *Why communication is important in data analytics lifecycle projects ?* **SPPU ; May-19, (End Sem), Marks 8**
5. *Demonstrate the overview of data analytics life cycle.* **SPPU : Oct.-19 (In Sem), Marks 5**

3.4 Multiple Choice Questions

Q.1 What are the different features of Big data analytics ?

a Open-source b Scalability

c Data recovery d All of above

Q.2 _______ data has internal structure but is not structured via pre-defined data models or schema.

a Structured b Semi-structured

c Unstructured d All of these

Q.3 In big data, ______ refer to heterogeneous sources and the nature of data, both structured and unstructured.

a volume b variety

c velocity d all of these

Q.4 Type of data analytics are ______.

a descriptive model b predictive model

c prescriptive model d all of these

Q.5 Data is collection of data objects and their ______.

a information b attributes

c characteristics d none

Q.6 Data frame is used for storing data _______.

a value
b numbers
c tables
d all of these

Q.7 Data frames is a collection of vectors that all have the _____ length.

a same
b variable
c short
d all of these

Q.8 Following which method is NOT used for handling missing values.

a Eliminate data objects
b Estimate missing values
c Ignore the missing value during analysis
d Replace with all error values

Q.9 Data _____ means removing the inconsistent data or noise and collecting necessary information of a collection of interrelated data.

a preprocessing
b cleaning
c transforming
d none

Q.10 The process of converting the integrating data into correct format is called _____.

a data cleaning
b data preprocessing
c data transformation
d data handling

Answer Keys for Multiple Choice Questions :

Q.1	d	**Q.2**	c
Q.3	b	**Q.4**	d
Q.5	b	**Q.6**	c
Q.7	a	**Q.8**	d
Q.9	b	**Q.10**	c

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Notes